

WELCOME!

E-Commerce (CSC370)

Instructor

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Course plan

- Syllabus
- Text book: Kenneth C. Laudon and Carol Guercio Traver, E-commerce Business Technology Society, Pearson
- Book freely available at:
[https://bawar.net/data0/books/5c63140ab1587/pdf/Kenneth_C_Laudon,_Carol_Guercio_Traver_-_E-Commerce_2017_\(2017,_Pearson\).pdf](https://bawar.net/data0/books/5c63140ab1587/pdf/Kenneth_C_Laudon,_Carol_Guercio_Traver_-_E-Commerce_2017_(2017,_Pearson).pdf)

Unit 1: Introduction

- E-commerce, E-business, Features of E-commerce, Pure vs. Partial E-commerce, History of E-commerce
- E-commerce Framework (People, Public Policy, Marketing and Advertisement, Support Services, Business Partnerships)
- Types of E-commerce: B2C, B2B, C2B, C2C,
- M-Commerce, U-commerce, Social-Ecommerce, Local E-commerce
- Challenges in E-commerce, Status of E-commerce in Nepal, Overview of Electronic Transaction Act of Nepal

E-commerce

- E-commerce is the process of buying and selling of information, products (goods) and services using electronic medium (usually over computer network).
- Transfer of traditional commerce to the computer network or Internet.
- In e-commerce we use electronic medium to perform commercial activities such as order, payment, and delivery.

E-business

- Basically, E-commerce and E-business are similar concepts. Some people view these things differently.
- Ecommerce is a narrow concept which includes only buying and selling of products and services. i.e. shopping by using electronic medium.
- **E-business** is a broad concept. E-business includes more activities than just buying and selling. E-business involves –
 - Transactions between business partners
 - Customer service
 - Collaboration
 - Internal processes to support buying, selling, hiring, planning etc.
- Hence, E-business is larger than E-commerce i.e. E-business includes E-commerce + other activities related to business.
- But mostly these two terms are used interchangeably.

Pure Vs Partial E-commerce

- Pure : All the aspects(activities) of commerce are digital (electronic)
- E.g. Amazon sells e-books by pure e-commerce
- Partial: Some (not all) activities are electronic. E.g. Restaurants deliver food to home with cash payment on delivery.

Order: electronic(online)

other activities : non-electronic

Features of E-commerce

- **Ubiquity** --- available everywhere
- **Global Reach**
- **24/7 service**
- **Easy to start new business**
- **Improved customer service**
- **Information richness**
- **Customization and personalization**

History of E-commerce

- The history of ecommerce started 40 years ago and, to this day, continues to grow with new technologies, innovations, and thousands of businesses entering the online market each year.
- Electronic Data Interchanges and teleshopping in the 1970s paved the way for the modern day ecommerce store.
- The history of ecommerce is closely intertwined with the history of the internet.
- Online shopping only became possible when the internet was opened to the public in 1991. Amazon.com was one of the first ecommerce sites in the US to start selling products online and thousands of businesses have followed since.

See text book **Kenneth C. Laudon and Carol Guercio Traver, E-commerce Business Technology Society, Pearson** for more details.

- Book freely available at:

[https://bawar.net/data0/books/5c63140ab1587/pdf/Kenneth_C_Laudon,_Carol_Guercio_Traver_-_E-Commerce_2017_\(2017,_Pearson\).pdf](https://bawar.net/data0/books/5c63140ab1587/pdf/Kenneth_C_Laudon,_Carol_Guercio_Traver_-_E-Commerce_2017_(2017,_Pearson).pdf)

Types of E-commerce

1. Business to consumer (B2C)

- This is most general type of e-commerce where consumers buy goods and services from businesses.
- For e.g. a customer buys a computer online from Dell company (dell.com)

2. Business to Business (B2B)

- In this type of e-commerce, both parties are businesses. e.g. Motorcycle manufacturing company “Hero” buys tyres from MRF company.

2. Consumer to Business (C2B)

- In this type of e-commerce, a consumer sells products and services to a business. e.g. a person sells old household items to a company which collects old items and reuses them. (e.g. we sell old iron and plastic to businesses which makes new items from old iron or plastic.)

2. Consumer to Consumer(C2C)

- In this type of e-commerce, both parties are consumers. e.g. there are many auction web sites where consumers can sell products to another consumer. An example of such website is olx.com where consumers can put their items on sale and another consumer buys it.

Mobile Computing and Commerce:

- Mobile Computing Technology
- Mobile Financial Services
- Mobile Shopping
- Advertisement
- Entertainment

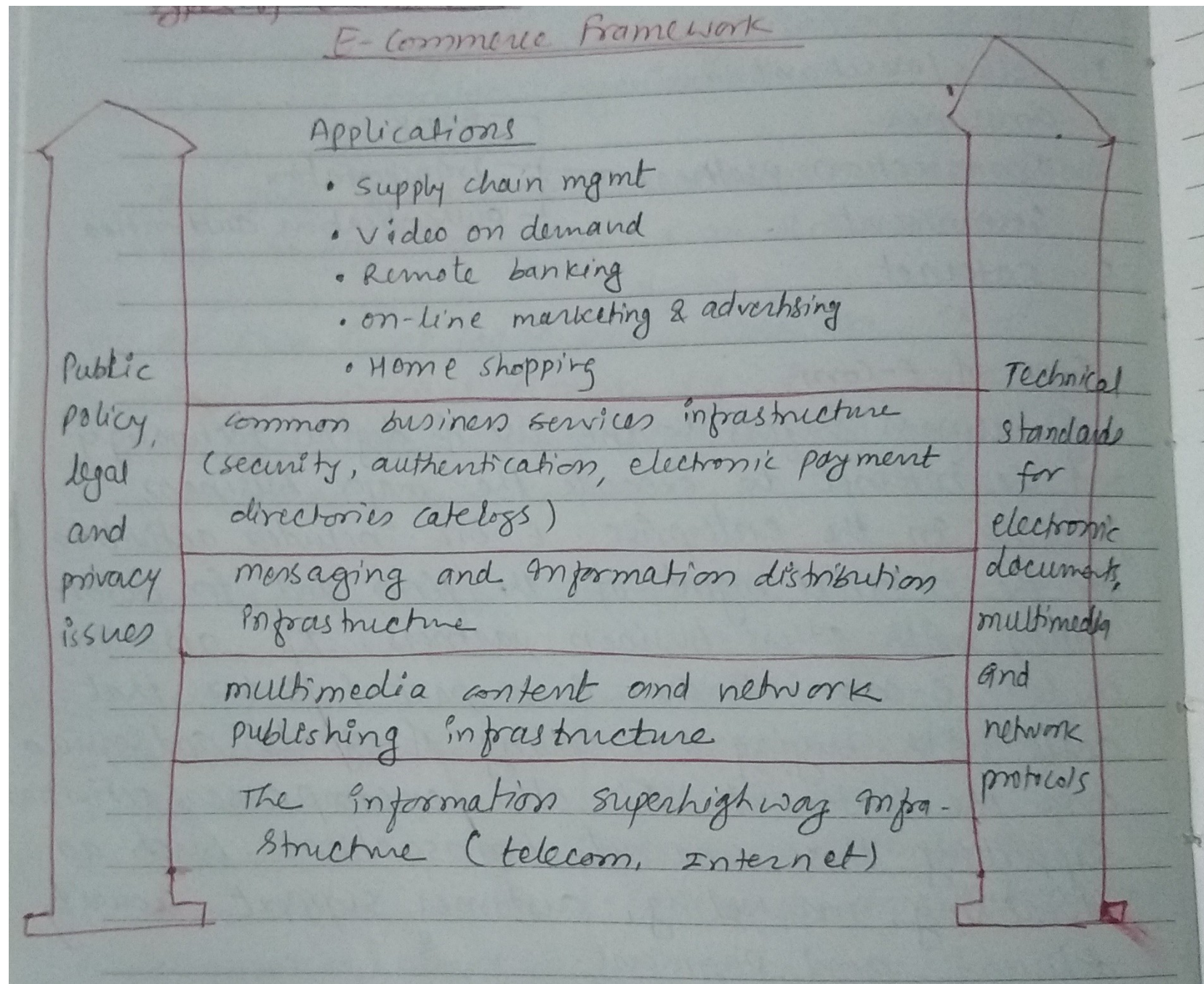
U-commerce

- **Ubiquitous Commerce** is also known as U-Commerce is the next generation of commerce.
- The ultimate form of e-**commerce** and m-**commerce** in an 'anytime, anywhere' fashion. It involves the use of **ubiquitous** networks to support personalized and uninterrupted communications and transactions at a level of value that far exceeds traditional **commerce**.
- Includes four major features:
 - *Ubiquitous* = represents the ability to be connect at any time and in any place as well as the integration of human-computer interaction into most devices and processes, e.g. household objects.
 - *Uniqueness* = stands for the unique identification of each customer or user regarding his identity, current context, needs and location resulting in an individual service.
 - *Universal* = is related to everyone's devices which can be used multifunctional and as well as universal -you will always be connected no matter of your place.
 - *Unison* = constitutes the data integration across applications and devices to provide users consistent and fully access to required information independent of device and location. The term unison also relates to fully synchronised devices at any time.

social commerce

- social commerce is the use of social network(s) in the context of e-commerce transactions.
- a subset of electronic commerce that involves social media, online media that supports social interaction, and user contributions to assist online buying and selling of products and services
- The term social commerce was introduced by Yahoo! in November 2005
- Some of the marketing tactics social commerce employs include:
 - Inviting users to vote on product style or choices
 - Offering personalized buyer options
 - Applying large and striking graphics to attract viewer clicks
 - Using videos to show the product in use and from multiple angles
 - Encouraging user-submitted photos, commentary, and feedback
 - Using celebrity endorsements of the product line
 - Linking directly to the checkout or shopping cart
 - Offering promotions or giveaways to users who share the product on their feeds

E-commerce Framework



Challenges in e-commerce

1. Economic challenges The economic challenges facing eCommerce merchants include—

- the costs related to establishing an eCommerce business,
- the number of competing online merchants,
- issues connected with infrastructure upgrades
- the availability (or shortage) of skilled staff. Researchers point out that it's estimated up to 90 percent of Internet host computers reside in high-income countries that are home to only 16 percent of the world's population.

Challenges in e-commerce

2. Technological challenges According to industry analysts, the primary technology-related challenges facing ecommerce businesses include

- security concerns
- bandwidth availability
- Reliability of network infrastructure
- integration with existing protocols

One of the realities of the explosive popularity of ecommerce is that it has also become a more attractive target for criminals; put simply, more overall ecommerce activity also means greater security concerns. In addition, with each new high profile ‘hacking’ incident, consumer confidence in online retailing takes a ‘hit’ and further raises the importance of ecommerce security. Additionally, another challenge of ecommerce booming is the growing concern about the reliability of network infrastructure. Integrating Internet software with preexisting applications and databases presents another technology-related challenge; ecommerce technology continuously evolves, and integrating that new technology is not always an easy—or inexpensive—task.

Challenges in e-commerce

3. Social challenges

- In recent years, consumers' concern over issues such as privacy, and sharing personal information has grown exponentially.
- Additionally, there are cultural differences that come into play when growing an ecommerce business beyond its native region. Unlike traditional merchants who were easily able to use language and marketing methods that played well with their target audience, online sellers have to take into consideration the fact that there are many linguistic, cultural and social differences in their extended customer base. Any one of those differences has the potential to create stumbling blocks for ecommerce companies in their efforts to develop long-term 'relationships' with a broad customer base.

Challenges in e-commerce

4. Legal challenges

- a relatively new industry with many unresolved legal issues –
 - pertaining to intellectual property rights
 - pertaining to taxation e.g. the recent US Supreme Court decision ruled that states can collect online sales taxes on businesses that are not physically located within their boundaries has the potential to dramatically impact the bottom line of ecommerce companies nationwide.

Status of E-commerce in Nepal

- Currently in Nepal, there are dozens of ISPs with millions of Internet users.
- Initially, E-Commerce Activities were mainly concentrated on Kathmandu Valley and some major cities.
- E-commerce was introduced in Nepal with an aim of letting Nepalis residing abroad send gifts to their families, friends, and relatives living here.
- The shift from physical stores to virtual ones began in the late 1990s.
- It is hard to find a reliable resource to pinpoint the start of ecommerce in Nepal. Many believe that thamel.com is the first-ever e-commerce company in Nepal. Thamel.com was started in 1998.

- Nowadays, with the increase of internet availability and the number of mobile users in rural areas, there is an increase in online activities.
- Today there are many E-Commerce Websites that are providing e-commerce e.g.
 - Daraz(daraz.com.np)
 - HamroBazar (hamrobazar.com)
 - NepBay (nepbay.com)
 - SastoDeal (sastodeal.com)
 - ESewaPasal(esewapasal.com)
 - Socheko.com, Smartdoko.com, gazabko.com, shopmandu.com, rojeko.com, merotarkari.com, merokirana.com etc...

Online resource: <https://www.everestparked.com/post/status-of-e-commerce-in-nepal#>

Challenges of ecommerce in Nepal

1. From buyer perspective

- Lack of trust
- Some people have discontinued buying online because of the bad experiences of getting worst customer service, delivery of not exact products, and no return policy.
- On top of that there is lack of versatile online payment gateway. We have some payment gateways such as esewa but they have limited services. Buyers buy from social media such as Facebook, Instagram but they can't pay online and are said to make cash payments when the goods will be delivered, i.e. cash on delivery
- Elder generation still prefer the traditional shopping habit, going from shops to shops and bargaining to the fullest and getting things done.

2. From Business perspective

- Lack of rules, regulations and government policy. Many international ecommerce companies are not operating in Nepal because of this.
- Unreliable networks.
- Lack of consumer awareness.

Electronic Transaction Act of Nepal

<http://www.lawcommission.gov.np/en/archives/16954>

- Enacted in 2063 (2008 AD)
- First attempt by Nepal to formally legalize and regulate electronic transactions.
- Objectives of the Electronic Transaction Act 2063
 - to make legal provisions for authentication and regularization of the recognition of electronic records
 - To make provisions for validity, integrity and reliability of generation, production, processing, storage, communication and transmission system of electronic records
 - To regulate all the relating matters of electronic transactions.
 - To control the acts of unauthorized use of electronic records or of making alteration in such records through the illegal manner

Scope of the Electronic Transaction Act 2063

- Creation and use of digital signature
- Control cyber/computer-related crimes.
- Protection of intellectual property.
- Protection of confidentiality.
- Regulations of an electronic transaction by establishing regulating bodies – such as
 - Office of Certificate Control (OCC)
 - Certificate Agencies (CA)
 - Subscribers

Major provisions of the Electronic Transaction Act 2063

1. Defines, regulates and recognizes electronic records and digital signature.
2. Defines the process of dispatch, receive an acknowledgment of electronic records.
3. It has the provision of few regulating bodies and their functions, rights, and duties such as –
 - Office of Certificate Control (OCC)
 - Certification Agencies (CA)
 - Subscriber

4. It has the provision relating the use of digital signature and certificate.
5. It has the provision relating to government use of the digital signature.
7. It has the provision relating to computer-related crimes and punishments.
8. Provision of IT tribunal is defined as the first jurisdictional and appellate jurisdiction.
IT tribunal

- First Jurisdictional
- Appellate Jurisdictional

9. It defines a few documents which this act is not applicable for

- All the negotiable instrument defined by the “Negotiable Instrument Act 2034”
- All the documents related to the ownership of properties (dhanipurja, laalpurja ect).
- Any documents which are used as a primary document in the court process

Reading resource: <https://www.ratojob.com/electronic-transaction-act/>

Unit 2

E-COMMERCE BUSINESS MODELS

2.1 Introduction

- A business model is a set of planned activities (sometimes referred to as business processes) designed to result in a profit in a marketplace.
- An e-commerce business model aims to use and leverage the unique qualities of the Internet, the Web, and the mobile platform.

KEY ELEMENTS OF A BUSINESS MODEL

- Whether it is traditional commerce or e-commerce, a successful business model must address following key elements:
 1. Value proposition
 2. Revenue model
 3. Market Opportunity
 4. Competitive environment
 5. Competitive advantage
 6. Market strategy
 7. Organizational development
 8. Management team.

1. Value proposition

- defines how a company's product or service fulfills the needs of customers
- To develop and/or analyze a firm's value proposition, we need to understand why customers will choose to do business with the firm instead of another company and what the firm provides that other firms do not and cannot.
- From the consumer point of view, successful e-commerce value propositions include personalization and customization of product offerings, reduction of product search costs, reduction of price discovery costs, and facilitation of transactions by managing product delivery.
- E.g. consider searching for/buying a book in traditional commerce Vs. Amazon

2. Revenue Model

- A firm's revenue model describes how the firm will earn revenue, generate profits, and produce a superior return on invested capital.
- Some ecommerce revenue models:
 - a) Advertising revenue mode
 - b) Subscription revenue model
 - c) Transaction fee revenue model
 - d) Sales revenue model
 - e) Affiliate revenue model

TABLE 2.2**FIVE PRIMARY REVENUE MODELS**

REVENUE MODEL	EXAMPLES	REVENUE SOURCE
Advertising	Yahoo	Fees from advertisers in exchange for advertisements
Subscription	eHarmony Consumer Reports Online Netflix	Fees from subscribers in exchange for access to content or services
Transaction Fee	eBay E*Trade	Fees (commissions) for enabling or executing a transaction
Sales	Amazon L.L.Bean Birchbox iTunes	Sales of goods, information, or services
Affiliate	MyPoints	Fees for business referrals

3. Market opportunity

- The term market opportunity refers to the company's intended marketspace (i.e., an area of actual or potential commercial value) and the overall potential financial opportunities available to the firm in that marketspace.

4. Competitive environment

- A firm's competitive environment refers to the other companies selling similar products and operating in the same marketspace. It also refers to the presence of substitute products and potential new entrants to the market, as well as the power of customers and suppliers over our business.

5. competitive advantage

- Competitive advantage is achieved by a firm when it can produce a superior product and/or bring the product to market at a lower price than most, or all, of its competitors

6. Market strategy

- Any business concept fails if it is not properly marketed to the potential customers.
- Marketing is the process of promoting and selling of products and services.
- E.g. Twitter and YouTube have a social network marketing strategy that encourages users to post their content for free, build personal profile pages, contact their friends, and build a community. In these cases, the customer becomes part of the marketing staff!

7. Organizational development

- It is the plan that describes how the company will organize the work that needs to be accomplished
- Typically, work is divided into functional departments, such as production, shipping, marketing, customer support, and finance. Jobs within these functional areas are defined, and then recruitment begins for specific job titles and responsibilities. Typically, in the beginning, generalists who can perform multiple tasks are hired. As the company grows, recruiting becomes more specialized.

8. Management team

- the single most important element of a business model is the management team responsible for making the model work. A strong management team gives a model instant credibility to outside investors, immediate market-specific knowledge, and experience in implementing business plans.

2.2 BUSINESS-TO-CONSUMER (B2C) BUSINESS MODELS

- **E-tailer**
- **Community Provider**
- **Content Provider**
- **Portal**
- **Transaction Broker**
- **Market Creator**
- **Service Provider**

- **E-tailer**

- E-tailers are online retail stores
- They are electronic versions of traditional brick-and-mortar stores. Hence e-tailers are referred to as brick-and-click stores.
- Customers can place an order and pay electronically
- May be new businesses or subsidiaries/ divisions of existing physical stores (e.g. Amazon, Walmart)

- **Community Provider**

- Community providers create an online environment where people with similar interests can transact (buy and sell goods); share interests, photos, videos; communicate with like-minded people; receive interest-related information; and even play out fantasies by adopting online personalities called avatars.
- Facebook, LinkedIn, Twitter, and Pinterest, and hundreds of other smaller, niche(specialized, small) social networks all offer users community-building tools and services.

- **Content Provider**

- Content providers distribute information content, such as digital video, music, photos, text, and artwork
- Content providers can make money via a variety of different revenue models, including advertising, subscription fees, and sales of digital goods.
- For instance, in the case of Apple Music, a monthly subscription fee provides users with access to millions of music tracks.

- **Portal**

- Portals offer users powerful search tools as well as an integrated package of content and services, such as news, e-mail, instant messaging, calendars, shopping, music downloads, video streaming, and more, all in one place.
- Examples: Yahoo, MSN
- Initially, portals sought to be viewed as “gateways” to the Internet. Today, however, the portal business model is to be a destination. They are marketed as places where consumers will hopefully stay a long time to read news, find entertainment, and meet other people (think of destination resorts).
- Portals generate revenue primarily by charging advertisers for ad placement, collecting referral fees for steering customers to other sites, and charging for premium services. They earn money from advertising etc.

- **Transaction Broker**

- Companies that process transactions for consumers normally handled in person, by phone, or by mail are transaction brokers.
- The largest industries using this model are financial services, travel services, and job placement services.
- The online transaction broker's primary value propositions are savings of money and time. In addition, most transaction brokers provide timely information and opinions.
- Transaction brokers make money each time a transaction occurs.
- e.g. Online stock brokers, Air ticket booking

- **Market Creator**

- Market creators build a digital environment in which buyers and sellers can meet, display and search for products and services, and establish prices.
- Prior to the Internet and the Web, market creators relied on physical places to establish a market.
 - Examples: Priceline -- allows consumers to set the price they are willing to pay for various travel accommodations and other products (sometimes referred to as a reverse auction).
 - eBay-- online auction site utilized by both businesses and consumers.
- Market creators make money by either charging a percentage of every transaction made, or charging merchants for access to the market. For example, eBay's auction business model is to create a digital environment for buyers and sellers to meet, agree on a price, and transact.
- How market creator model is different from transaction broker model

- **Service Provider**

- service providers offer services (not products as in e-tailer model) online.
- Recently, there's been an explosion in online services.
- Photo sharing, video sharing, and user-generated content (in blogs and social networks)
- Google has led the way in developing online applications such as Google Maps, Google Docs, and Gmail.
- Other personal services such as online medical bill management, financial and pension planning, and travel recommendation are showing strong growth.
- Revenue ? –
 - Some charge a fee, or monthly subscriptions
 - Through advertising and by collecting personal information that is useful in direct marketing.
 - Many service providers employ a freemium revenue model, in which some basic services are free, but others require the payment of additional charges.
- value proposition? –
 - offer consumers valuable, convenient, time-saving, and low-cost alternatives to traditional service providers or provide services that are truly unique.

BUSINESS MODEL	VARIATIONS	EXAMPLES	DESCRIPTION	REVENUE MODELS
E-tailer	Virtual Merchant	Amazon Blue Nile Bluefly	Online version of retail store, where customers can shop at any hour of the day or night without leaving their home or office	Sales of goods
	Bricks-and-Clicks	Walmart Sears	Online distribution channel for a company that also has physical stores	Sales of goods
	Catalog Merchant	L.L.Bean LillianVernon	Online version of direct mail catalog	Sales of goods
	Manufacturer-Direct	Dell Mattel	Manufacturer uses online channel to sell direct to customer	Sales of goods
Community Provider		Facebook LinkedIn Twitter Pinterest	Sites where individuals with particular interests, hobbies, common experiences, or social networks can come together and "meet" online	Advertising, subscription, affiliate referral fees
Content Provider		Wall Street Journal CNN ESPN Netflix	Offers customers newspapers, magazines, books, film, television, music, games, and other forms of online content	Advertising, subscription fees, sales of digital goods

Portal	Horizontal/ General	Yahoo AOL MSN Facebook	Offers an integrated package of content, search, and social network services: news, e-mail, chat, music downloads, video streaming, calendars, etc. Seeks to be a user's home base	Advertising, subscription fees, transaction fees
	Vertical/ Specialized (Vortal)	Sailnet	Offers services and products to specialized marketplace	Advertising, subscription fees, transaction fees
	Search	Google Bing Ask	Focuses primarily on offering search services	Advertising, affiliate referral
Transaction Broker		E*Trade Expedia Monster Travelocity Orbitz	Processors of online sales transactions, such as stockbrokers and travel agents, that increase customers' productivity by helping them get things done faster and more cheaply	Transaction fees
Market Creator		eBay Etsy Amazon Priceline	Businesses that use Internet technology to create markets that bring buyers and sellers together	Transaction fees
Service Provider		VisaNow Wave RocketLawyer	Companies that make money by selling users a service, rather than a product	Sales of services

BUSINESS-TO-BUSINESS (B2B) BUSINESS MODELS

- Business-to-business (B2B) e-commerce, in which businesses sell to other businesses, is actually larger (more than 10 times) than the size of B2C e-commerce

BUSINESS MODEL	EXAMPLES	DESCRIPTION	REVENUE MODEL
(1) NET MARKETPLACE			
E-distributor	Grainger Amazon Business	Single-firm online version of retail and wholesale store; supply maintenance, repair, operation goods; indirect inputs	Sales of goods
E-procurement	Ariba Supplier Network PerfectCommerce	Single firm creating digital markets where sellers and buyers transact for indirect inputs	Fees for market-making services, supply chain management, and fulfillment services
Exchange	Go2Paper	Independently owned vertical digital marketplace for direct inputs	Fees and commissions on transactions
Industry Consortium	TheSeam SupplyOn	Industry-owned vertical digital market open to select suppliers	Fees and commissions on transactions
(2) PRIVATE INDUSTRIAL NETWORK			
	Walmart Procter & Gamble	Company-owned network that coordinates supply chains with a limited set of partners	Cost absorbed by network owner and recovered through production and distribution efficiencies

1. E-Distributor

- Companies that supply products and services directly to individual businesses are e-distributors.

2. E-Procurement

- **Procurement** is the process of finding and agreeing to terms, and acquiring goods or services.
- e-procurement firms create and sell access to digital markets.
- E.g. Ariba have created software that helps large firms organize their procurement process
 - Ariba creates custom-integrated online catalogs (where supplier firms can list their offerings) for purchasing firms.
 - On the sell side, Ariba helps vendors sell to large purchasers by providing software to handle catalog creation, shipping, insurance, and finance.
- Procurement software is referred to generically as “value chain management” software.

3. Exchange

- An exchange is an independent digital marketplace where hundreds of suppliers meet a smaller number of very large commercial purchasers.
- Exchanges are owned by independent, usually entrepreneurial startup firms whose business is making a market, and they generate revenue by charging a commission or fee based on the size of the transactions conducted among trading parties.
- They focus on the exchange of direct inputs to production and short-term contracts or spot purchasing.
- For buyers, B2B exchanges make it possible to gather information, check out suppliers, collect prices, and keep up to date on the latest happenings all in one place.
- Sellers benefit from expanded access to buyers.

4. Industry consortium

- Industry consortia are industry-owned vertical marketplaces that serve specific industries, such as the automobile, aerospace, chemical etc.
 - Vertical marketplaces: supply a smaller number of companies with products and services of specific interest to their industry
 - Horizontal marketplaces:
 - supply companies in different industries with a particular type of product and service.
 - sell specific products and services to a wide range of companies.
- For example, SupplyOn, owned by industrial giants Bosch (one of the world's largest suppliers of automotive components), Continental (a leading automotive manufacturing company), and Schaeffler (a global manufacturer of various types of bearings), provides a shared supply chain collaboration platform for companies in various manufacturing industries.

5. **Private Industrial network**

- sometimes referred to as a private trading exchange or PTX
- It is a digital network designed to coordinate the flow of communications among firms engaged in business together. The network is owned by a single large purchasing firm.
- Participation is by invitation only to trusted long-term suppliers of direct inputs.
- For instance, Walmart operates one of the largest private industrial networks in the world for its suppliers, who on a daily basis use Walmart's network to monitor the sales of their goods, the status of shipments, and the actual inventory level of their goods.

Electronic Data Interchange (EDI)

- a communications standard for sharing business documents and settlement information among a small number of firms.
- first developed in the mid-1970s and 1980s.
- a set of standards for structuring information that is to be electronically exchanged between and within businesses, organizations, government entities and other groups, without human intervention.
- An inter-company, application-to-application communication of data in standard format for business transactions.
- EDI is a broadly defined communications protocol for exchanging documents among computers using technical standards developed by the American National Standards Institute (ANSI X12 standards) and international bodies such as the United Nations (EDIFACT standards).

- EDI was developed to reduce the cost, delays, and errors inherent in the manual exchanges of documents such as purchase orders, shipping documents, price lists, payments, and customer data.
- EDI differs from an unstructured message because its messages are organized with distinct fields for each of the important pieces of information in a commercial transaction such as transaction date, product purchased, amount, sender's name, address, and recipient's name.
- EDI remains very important in the development of B2B e-commerce as EDI transactions total trillion dollars (about 50% of all B2B commerce)

EDI layered architecture:-

EDI semantic layer	Application Layer Services	
EDI standard layer	EDIFACT business form standards	
	ANSI X12 business form standards	
EDI transport layer	Electronic mail	X.435, MIME
	Point to point	FTP, TELNET
	World Wide Web	HTTP
Physical layer	Dial-up lines, Internet, I-way	

1) Semantic layer

- It describes the business application that is driving EDI.
- For a procurement application, this translates into requests for quotes, price quotes, purchase orders, acknowledgements.
- The information seen at this layer must be translated from a company specific form to a more generic form so that it can be send to various trading partners, who could be using a variety of software applications at this end.
 - When a trading partner sends a document, the EDI translation software converts the proprietary format into a standard mutually agreed on by the processing system.
 - When a company receivers the document, their EDI translation software automatically changes the standard format into proprietary format of their document processing software so that company can manipulate the information in whatever way it chooses to.

2. EDI standards layer

- It specifies business form structure and it also influences the content at application layer.
- set of standards and rules that determine a specific syntax for structuring and transferring electronically business documents between partners. There are different types of business documents in EDI, which are often called “Transaction Sets”.
- The Transaction Set – is a single business document such as a Purchase Order, Invoice, Student Educational Record (Transcript), Healthcare claim, Ship Notice.
- The most two important standards are:-
 - EDIFACT – European standard
 - ANSI X12 – American Standard

Refer to online resources for details of these standards. One source is :
<https://www.edi2xml.com/blog/ansi-asc-x12-standards-overview/>

3.EDI transport layer

- This layer is responsible for transfer of documents from sender to receiver.
- This layer can be viewed as application layer of TCP/IP (Internet) when we use Internet as physical layer. In such case various TCP/IP application layer protocols can be utilized as transport layer of EDI.

4.Physical layer

- Physical layer is the medium of communication over a network.
- This layer includes various media such as Dial up lines (old technology), Internet, Private networks etc.

When to use EDI?

- EDI is an important industrial network technology, suited to support communications among a small set of strategic partners in direct, long-term trading relationships.

Cons(weaknesses) of EDI?

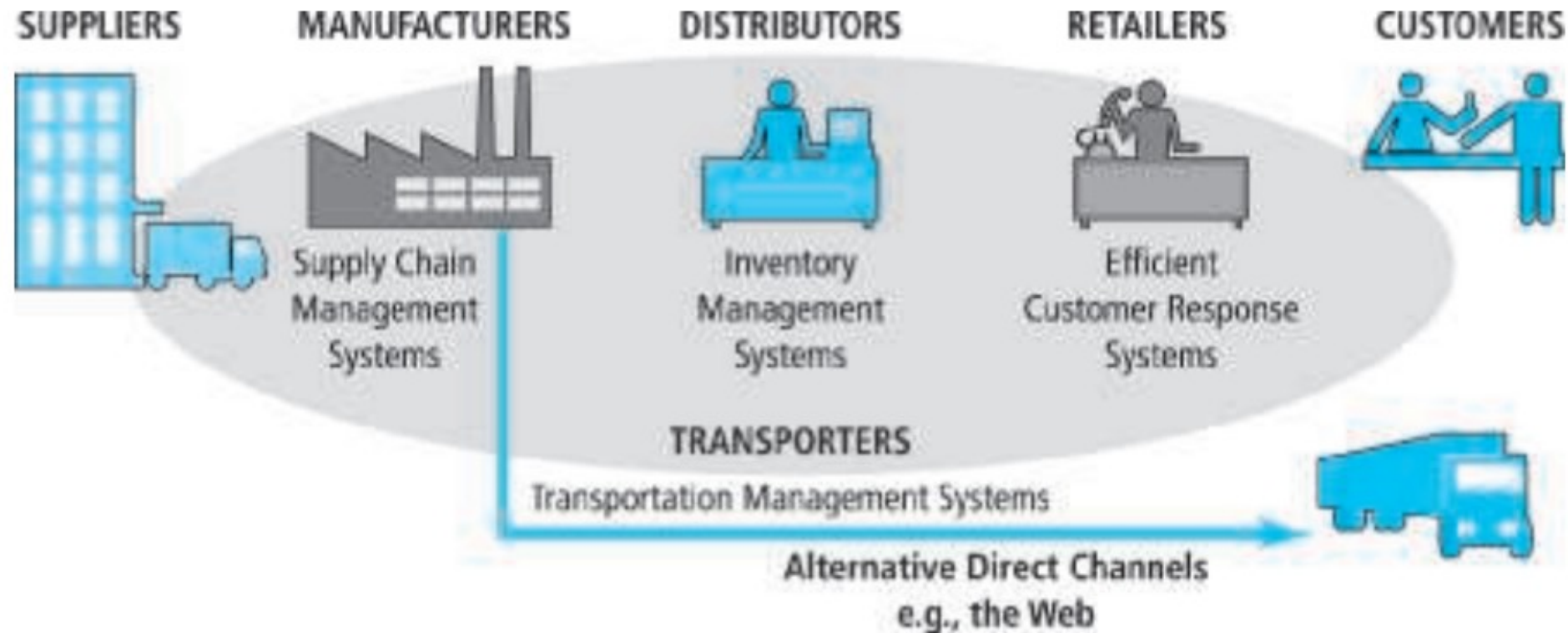
- EDI is not well suited for the development of Net marketplaces, where thousands of suppliers and purchasers meet in a digital arena to negotiate prices.
- EDI supports direct bilateral communications among a small set of firms and does not permit the multilateral, dynamic relationships of a true marketplace.
- EDI does not provide for price transparency among a large number of suppliers, does not scale easily to include new participants, and is not a real-time communications environment.
- EDI does not have a rich communications environment that can simultaneously support e-mail messaging, video conferencing, sharing of graphic documents, network meetings, or user-friendly flexible database creation and management

E-commerce and Industry value chain

- A value chain is the set of activities performed in an industry or in a firm that transforms raw inputs into final products and services. Each of these activities adds economic value to the final product; hence, the term value chain as an interconnected set of value adding activities

FIGURE 2.4

E-COMMERCE AND INDUSTRY VALUE CHAINS



- Above figure illustrates the six generic players in an industry value chain: suppliers, manufacturers, transporters, distributors, retailers, and customers.
- One of the basic tools for understanding the impact of information technology on industry and firm operations is the value chain.
- A detailed industry value chain analysis can help identify more precisely just how e-commerce may change business operations at the industry level.
- Every industry can be characterized by a set of value-adding activities performed by a variety of actors.
- E-commerce potentially affects the capabilities of each player as well as the overall operational efficiency of the industry.

Impact of E-commerce on Industry value chain

- By reducing the cost of information, e-commerce offers each of the key players in an industry value chain new opportunities to maximize their positions by lowering costs and/or raising prices.
- **Manufacturers** can reduce the costs they pay for goods by developing Internet-based B2B exchanges with their suppliers.
- **Manufacturers** can develop direct relationships with their customers, bypassing the costs of distributors and retailers.
- **Distributors** can develop highly efficient inventory management systems to reduce their costs, and retailers can develop highly efficient customer relationship management systems to strengthen their service to customers.
- **Customers** in turn can search for the best quality, fastest delivery, and lowest prices, thereby lowering their transaction costs and reducing prices they pay for final goods.
- The operational efficiency of the entire industry can increase, lowering prices and adding value for consumers, and helping the industry to compete with alternative industries.

FIRM VALUE CHAINS

- The concept of value chain can be used to analyze a single firm's operational efficiency as well.
- A firm value chain is the set of activities a firm engages in to create final products from raw inputs. Each step in the process of production adds value to the final product.

How does e-commerce technology potentially affect the value chains of firms within an industry?

- Firms develop support activities that coordinate the production process and contribute to overall operational efficiency.
- Following figure illustrates the key steps and support activities in a firm's value chain.

FIGURE 2.5

E-COMMERCE AND FIRM VALUE CHAINS

Administration
Human Resources
Information Systems
Procurement
Finance/Accounting

SECONDARY ACTIVITIES

PRIMARY ACTIVITIES



Inbound
Logistics



Operations



Outbound
Logistics



Sales and
Marketing



After Sales
Service

- E-commerce offers firms many opportunities to increase their operational efficiency and differentiate their products.
 - Firms can use the Internet's communications efficiency to outsource some primary and secondary activities to specialized, more efficient providers without such outsourcing being visible to the consumer.
 - Firms can use e-commerce to more precisely coordinate the steps in the value chains and reduce their costs.
 - Firms can use e-commerce to provide users with more differentiated and high-value products.
 - For instance, Amazon provides consumers with a much larger inventory of books to choose from, at a lower cost, than traditional book stores. It also provides many services—such as instantly available professional and consumer reviews, and information on buying patterns of other consumers—that traditional bookstores cannot.

FIRM VALUE WEBS

- While firms produce value through their value chains, they also rely on the value chains of their partners—their suppliers, distributors, and delivery firms.
- A value web is a networked business ecosystem that uses e-commerce technology to coordinate the value chains of business partners within an industry, or at the first level, to coordinate the value chains of a group of firms.
- Following Figure illustrates a value web.

FIGURE 2.6**INTERNET-ENABLED VALUE WEB**

- E-commerce creates new opportunities for firms to cooperate and create a value web.
- A value web coordinates a firm's suppliers with its own production needs using an Internet-based supply chain management system.
- Firms use the Internet to develop close relationships with their logistics partners.
- Example:
 - Amazon relies on UPS tracking systems to provide its customers with online package tracking, and it relies on the U.S. Postal Service systems to insert packages directly into the mail stream.
 - Amazon has partnership relations with hundreds of firms to generate customers and to manage relationships with customers.
 - The value Amazon delivers to customers is in large part the result of coordination with other firms and not simply the result of activities internal to Amazon. The value of Amazon is, in large part, the value delivered by its value web partners.

Case Study

Read yourself the cases in the text book.